Arithmetic Quantum Cat Maps:
Exact Moment Reductions, Finite Certificates, and the Remaining BGS Obstruction

Maksim Churilov

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Abstract

We give a conservative reconstruction of a computational research programme on arithmetic quantum cat maps. The main verified contribution is an exact pair-key reduction for the fourth-moment geometry. It yields the general constrained-pair count $(K-1)(K-2)$, isolates a weighted diagonal key, decomposes the fourth-moment difference distribution into off/off, mixed, and diagonal pieces, and turns the mixed term through a Cayley transform into an additive-energy problem. An exhaustive integer matching audit and exact cyclotomic resultants give

$$E_{11} = \{23, 67, 89\}, \qquad E_{13} = \{53, 79, 131, 157, 313\},$$

while the exact design $(p, K) = (461, 23)$ has collision energy 1190 against a forced baseline 902, ruling out a naive uniform-in-K extrapolation. The remaining off/off term has a promising elliptic-family organization, but the required projective, bad-fiber, sheaf, isogeny and boundary arguments are not yet closed; its square-root correlation estimate is therefore a theorem draft, not a theorem. Earlier finite Lean certificates, twisted-Jacobi regressions, scoped antiunitary symmetry, and corrected finite GOE diagnostics are also recorded at their actual evidence levels. No result here proves microscopic Bohigas–Giannoni–Schmit (BGS) eigenphase statistics.

1 Purpose, conventions, and evidence levels

The historical workspace mixed exact finite identities, useful numerical experiments, unfinished proof routes, and several statements labelled "proved" too early. The present manuscript is the publication-facing description of the part that survives an adversarial reconstruction audit.

Throughout, p is an odd prime and $H \leq \mathbf{F}_p^\times$ is a multiplicative subgroup of order K. Division means multiplication by the inverse in $\mathbf{F}_p$. We use the following evidence levels:

- **Exact**: a current algebraic derivation with stated hypotheses.
- **Finite certificate**: an exhaustive or kernel-checked proposition at explicit finite parameters.
- **Finite regression**: an implementation/reference agreement at explicit finite parameters.
- **Numerical**: descriptive floating-point evidence only.
- **Draft**: a proof architecture with named unresolved obligations.

- **Open**: the stated closure condition has not been achieved.

The distinction between moment information and microscopic spectral statistics is load-bearing. Established cat-map literature includes Hecke-eigenfunction results, matrix-element moments and quantum ergodicity [1, 2, 5, 6], as well as higher-dimensional eigenfunction delocalization/semiclassical-measure restrictions [4]. These do not by themselves provide a microscopic eigenphase local law.

2 Exact pair-key geometry

For $x, y \in \mathbf{F}_p \setminus \{0, 1\}$ define

$$u = \frac{x}{y}, \qquad v = \frac{x-1}{y-1}, \qquad d = x - y. \tag{1}$$

Proposition 2.1 (Off-diagonal pair-key bijection)**.** *If $x \neq y$, then $u, v \in \mathbf{F}_p^\times \setminus \{1\}$, $u \neq v$, and*

$$y = \frac{1-v}{u-v}, \qquad x = \frac{u(1-v)}{u-v}, \qquad d = \frac{(u-1)(v-1)}{v-u}. \tag{2}$$

Conversely every pair $u, v \in \mathbf{F}_p^\times \setminus \{1\}$ with $u \neq v$ arises from exactly one off-diagonal pair (x, y).

Proof. From $x = uy$ and $x - 1 = v(y - 1)$ we obtain $(u - v)y = 1 - v$, which gives the first two formulas in (2). Then

$$x - y = (u-1)y = \frac{(u-1)(1-v)}{u-v} = \frac{(u-1)(v-1)}{v-u}.$$

The inverse formulas prove both injectivity and surjectivity. □

The diagonal is exceptional: all $p - 2$ pairs $x = y \in \mathbf{F}_p \setminus \{0, 1\}$ collapse to

$$(u, v, d) = (1, 1, 0)$$

with multiplicity $p - 2$. This multiplicity is responsible for several historical factor errors.

Corollary 2.2 (Exact constrained-pair count)**.** *Suppose the two pair constraints require $u, v \in H$. The number of admissible off-diagonal ordered pairs is*

$$(K-1)(K-2).$$

Proof. By Proposition 2.1, constrained off-diagonal pairs are in bijection with ordered pairs $(u, v) \in (H \setminus \{1\})^2$ satisfying $u \neq v$. There are $(K - 1)(K - 2)$ of them. □

Status: Exact.. A Lean certificate independently checks the parametrization and count at $(p, K) = (53, 13)$; the finite certificate is a regression for, not the source of, the general proof.

3 Fourth-moment coupling and the exact distribution

Give each collapsed key $r = (u_r, v_r, d_r)$ a multiplicity n_r, equal to one off the diagonal and $p - 2$ at $(1, 1, 0)$. Define

$$B(t) = \sum_{r,s} n_r n_s \, \mathbf{1}_H(u_r u_s) \, \mathbf{1}_H(v_r v_s) \, \mathbf{1}_{d_r + d_s = t}. \tag{3}$$

Let $O(t)$ be the off/off contribution and set

$$C_H(t) = \#\left\{(u,v) \in (H \setminus \{1\})^2 : u \neq v,\ \frac{(u-1)(v-1)}{v-u} = t\right\}.$$

The exact diagonal bookkeeping is

$$B(t) = O(t) + 2(p-2)C_H(t) + (p-2)^2 \mathbf{1}_{t=0}. \tag{4}$$

The factor two is forced because diagonal/off and off/diagonal are distinct ordered classes.

Proposition 3.1 (Exact total coupled count)**.** *For the corrected pair-key model,*

$$|V_4| = (K^2+1)p^2 - (6K^2+3K)p + 13K^2 - 3K. \tag{5}$$

Status: Exact, with independent finite checks.. The maintained Python and C++ implementations reproduce, among others,

$$|V_4| = 48621 \quad (p,K) = (23,11), \qquad |V_4| = 423879 \quad (p,K) = (53,13),$$

and the Lean fourth-moment certificate independently checks the second design.

4 Character orthogonality and exact Fourier moments

Let

$$S = H^{\perp}$$

be the multiplicative-character annihilator of H. Then

$$m = |S| = \frac{p-1}{K}.$$

For the maintained twisted-Jacobi convention,

$$\sum_{\chi,\psi \in S} |J(\chi,\psi;q)|^4 = m^2 \Sigma_4(q), \qquad \Sigma_4(q) = \sum_{t \in \mathbf{F}_p} B(t) e_p(qt), \tag{6}$$

where $e_p(z) = \exp(2\pi i z/p)$. Reversing the Fourier sign only conjugates the transform and does not alter the mean/variance statements below.

Write

$$V = \sum_t B(t), \qquad Z = B(0).$$

For q uniformly distributed over $\mathbf{F}_p^{\times}$,

$$\mathbb{E}_{q \neq 0} \Sigma_4(q) = \frac{pZ - V}{p-1}. \tag{7}$$

Moreover,

$$\mathrm{Var}_{q \neq 0}\, \Sigma_4(q) = p\ \mathrm{Var}_{t \neq 0}\, B(t). \tag{8}$$

Proof. For $t \in \mathbf{F}_p$,

$$\sum_{q \neq 0} e_p(qt) = \begin{cases} p-1, & t = 0, \\ -1, & t \neq 0. \end{cases}$$

This gives (7). Applying finite Parseval to $\Sigma_4(q) = \widehat{B}(q)$ and subtracting the nonzero-frequency mean gives (8). □

Thus the fluctuation problem is a concrete finite L^2 problem for B, not a probabilistic independence heuristic.

5 The mixed term as Cayley additive energy

Introduce

$$c(h) = \frac{h+1}{h-1}, \qquad h \in H \setminus \{1\}.$$

For $u \neq v$,

$$c(u) - c(v) = \frac{2(v-u)}{(u-1)(v-1)},$$

so

$$d(u,v) = \frac{2}{c(u)-c(v)}. \tag{9}$$

Let $A = c(H \setminus \{1\})$. Since $x \mapsto 2/x$ permutes $\mathbf{F}_p^\times$, the nonzero multiplicities of C_H are the nonzero difference multiplicities of A. Consequently

$$\sum_t C_H(t)^2 = E_+(A) - |A|^2, \qquad E_+(A) = \sum_x r_{A-A}(x)^2. \tag{10}$$

Also $c(h^{-1}) = -c(h)$, so $A = -A$ and $0 \notin A$. The forced central-symmetry collision baseline used by the finite verifier is

$$(K-1)(2K-5). \tag{11}$$

The same representation count has an exact PGL_2 intersection interpretation. This structural formulation replaces the historical Möbius-fiber argument, whose $d = 0$ fiber is degenerate.

6 Exact matching audit and fixed-K exceptional primes

Clearing a collision between two Cayley differences and matching the six terms gives $6! = 720$ integer linear systems.

Proposition 6.1 (Computer-assisted matching classification)**.** *The exact verifier classifies the* 720 *systems as*

$$30 \text{ of rank } 2, \qquad 240 \text{ of rank } 3, \qquad 450 \text{ of rank } 4$$

over $\mathbb{Q}$*. The gcds of maximal nonzero minors show that an additional rank drop can occur only in characteristic* 2*. The admissible positive-dimensional low-rank systems are the identity and inversion-swap symmetries; the remaining low-rank kernels force forbidden zero/diagonal configurations.*

Status: Exact exhaustive finite algebra.. The integer matrices are recomputed in CI.

For a specified odd prime K, a normalized non-forced collision produces a cyclotomic algebraic integer. If it is nonzero, a reduction collision can occur only at rational primes dividing its norm. The current repository performs the complete normalized nonvanishing/resultant enumeration for $K = 11$ and $K = 13$.

Theorem 6.2 (Audited exceptional sets)**.** *The exact computer-assisted resultant computation gives*

$$E_{11} = \{23, 67, 89\}, \qquad E_{13} = \{53, 79, 131, 157, 313\}.$$

For $K = 11$ *the audit checks* 793 *normalized non-forced quadruples and finds* 8 *distinct nonzero norms; for* $K = 13$ *it checks* 1431 *quadruples and finds* 14 *distinct nonzero norms.*

Status: Finite/computer-assisted exact certificate for the two stated values of K.. Representative finite reductions test both directions:

(p, K)	predicted exceptional?	$\sum_t C_H(t)^2 -$ baseline	result
$(53, 13)$	yes	208	excess observed
$(67, 11)$	yes	48	excess observed
$(199, 11)$	no	0	forced only
$(521, 13)$	no	0	forced only.

6.1 Non-uniformity witness

At $(p, K) = (461, 23)$,

$$\sum_t C_H(t)^2 = 1190,$$

while (11) is

$$22(46 - 5) = 902.$$

The exact excess is 288. This does not contradict the fixed-K mechanism; it blocks a silent extrapolation from $K = 11, 13$ to a useful uniform-in-growing-K formula.

7 The off/off elliptic route

After the mixed term is separated, the unresolved contribution is the centered energy of $O(t)$. Historical symbolic computations organize the relevant fibers as a one-parameter genus-one family and suggest a quartic/Weierstrass model, a nonconstant rational j-map, visible bad fibers, and same-j parameter symmetries.

Status: Theorem draft; historical symbolic structure is not independently re-generated by current CI.. A publication-grade proof must re-establish or certify the structural calculation and then verify:

1. a smooth projective model over an explicit open base;
2. the complete discriminant and bad-fiber locus, including characteristic exceptions;
3. the affine/projective boundary correction;
4. purity, lissity, geometric irreducibility, and uniform conductor/ramification control for the actual R^1 sheaf;
5. every relevant geometric isomorphism and twist between parameter fibers;
6. separable and inseparable isogeny exclusions or exceptional strata;
7. the exact hypotheses of the chosen trace-sheaf correlation theorem.

Michel-type family arguments [7], quantitative sheaf theory [8], Tate/isogeny technology [9], and the modern joint-family example of Grove–Saad [10] are relevant tools. None is a plug-in proof for this pencil.

Conditional on the required square-root correlation estimate with uniform constants, the existing combinatorics lead to

$$O\Big(K^2 p + K^4 p^{1/2} + K^4\Big), \tag{12}$$

with the intended restricted regime $K = o(p^{1/4})$. Equation (12) remains a draft conclusion.

8 Earlier results retained at their safe scope

8.1 Finite torus geometry at $p = 199$

The active Lean module `TorusLevelSets.lean` verifies for

$$Q(a, b) = a^2 - ab - b^2 \pmod{199}$$

and the action of $A^T = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$: the generator has order 11; each nonzero level contains 198 nonzero points; the zero cone minus the origin has 396 points; each nonzero level has 18 cyclic orbits; and the cyclic action is free there. These are fixed finite `native_decide` certificates, not a theorem for all primes.

8.2 Finite branch scale at $p = 199$

`TorusScaleLaw.lean` checks that $a = 139$ has order 11, its recorded inverse is 63, and the branch scales 123 and $76 = -123$ satisfy

$$123^2 \equiv 76^2 \equiv 5, \qquad 123 \cdot 76 \equiv -5 \pmod{199}.$$

Again this is a design certificate, not a general scale law.

8.3 Twisted-Jacobi finite regressions

Maintained scalar-versus-vectorized checks and preserved q-curve records validate the implementation at the explicit designs. A historical formula omitted a domain condition: the relevant n_1 must be nonzero. That condition is automatic on the audited nonzero-q curves, so the finite regression survives while the unrestricted historical statement does not.

8.4 Antiunitary symmetry: safe statement

If an antiunitary Θ satisfies $\Theta U \Theta^{-1} = U^{-1}$ and $Uv = \lambda v$ with $|\lambda| = 1$, then anti-linearity gives

$$U\Theta v = \overline{\lambda}^{-1}\Theta v = \lambda \Theta v.$$

Thus the eigenspace is preserved. We do not promote a blanket theorem that every relevant global block has already been certified as GOE class; that would require a convention-consistent treatment of all kernel charts and desymmetrizations if it becomes load-bearing.

8.5 Shape/Catalan, arcsine, and higher-rank historical material

Historical exact finite relation counts and finite moment patterns motivated Catalan/arcsine conjectures, but the old word-trace reduction has an unresolved higher rewrite case. Finitely many matching moments do not prove an all-orders law. Likewise historical higher-rank/$\mathrm{Sp}(2g)$ diagonal and coset diagnostics remain useful test cases in Git history, but no general higher-rank theorem is promoted from them.

9 Corrected finite spectral diagnostics

The finite spectral records are deliberately descriptive. They use the full complex Hermitian sector matrix

$$\mathcal{H}(q) = M(q) + M(q)^*.$$

A reconstruction-era implementation accidentally diagonalized $\mathrm{Re}\,\mathcal{H}(q)$ instead. Hermitian matrices need not be entrywise real, so this changed the finite spectra. The bug was found by an independent comparison, a regression guard now separates the two spectra, and the promoted numerical records were regenerated. No exact result above depended on those records.

For the corrected spacing diagnostic, polynomially unfolded gaps are pooled only to report the empirical CDF sup-distance to the GOE Wigner surmise:

p	K	sector dimension	$\#q$	D_{CDF}
3571	17	105	40	0.0077766786
19489	29	336	43	0.0087662249.

No IID Kolmogorov–Smirnov significance is attached because the unfolded gaps are dependent.

The second diagnostic uses forward windows anchored at each eigenvalue and truncated at the spectral edge. It is *not* the stationary number variance $\Sigma^2(L)$. For $(p, K) = (19489, 29)$, 25 cat-map samples and 60 GOE simulations with fixed seed 20260901 give

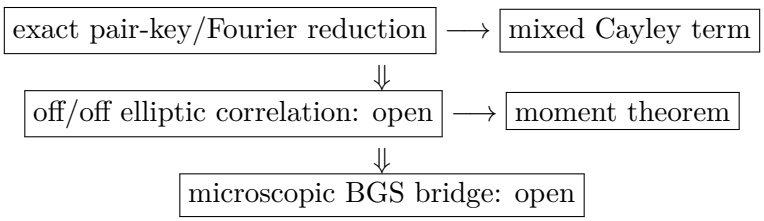
L	cat-map mean	GOE mean	descriptive standardized difference
1	0.328504	0.340868	2.3833
2	0.492778	0.508477	1.5774
3	0.578620	0.609855	2.3600
5	0.778921	0.791360	0.6510
8	1.203787	1.243551	2.0698
12	2.416266	2.465283	2.2076

The standardized column is only a scale for the matched finite comparison; it is not a Gaussian hypothesis-test statistic. These observations are numerical context, compatible with earlier cat-map spacing experiments [3], not evidence for a limiting BGS theorem.

10 Why microscopic BGS remains open

A BGS statement concerns local eigenphase correlations at the mean-spacing scale after relevant symmetries are removed. The exact pair-key identities and even the prospective estimate (12) are moment/fluctuation statements about a structured deterministic arithmetic family.

The current logical chain is therefore



Closing an earlier box does not close a later one. A local-statistics result still needs an explicit comparison/local-law mechanism (or a genuinely arithmetic replacement) with its scale, dependence structure and symmetry assumptions proved for the actual family.

11 Reproducibility, language migration, and negative controls

The maintained repository has one current implementation role per language:

- Python for checked finite-field/reference/symbolic orchestration and numerical diagnostics;
- three C++20 programs for independent exact/exhaustive finite checks: pair-key fourth moments, Cayley energy, and cat-map order;
- six active Lean 4 modules for finite certificates and a historical falsifier;
- TeX plus a committed synchronized PDF for the manuscript.

The current branch contains no Julia source. All 31 historical Julia paths (23 byte-distinct sources) have final dispositions: useful current logic was replaced by maintained C++/Python where warranted, while false, duplicate or non-load-bearing code remains recoverable in Git history rather than being ported mechanically.

CI checks whole-repository structure, promoted-evidence provenance, Markdown links, Python syntax warnings, exact Python regressions, complete committed numerical diagnostics, all maintained C++ programs, the Lean import graph and `lake build`, and an independent manuscript build. The fixed-K Cayley JSON is regenerated from its exact producer and compared with the committed record.

Negative controls are retained when they test a different failure mode: the degenerate historical Möbius fiber is a Lean finite falsifier, and $(461, 23)$ is an independent Python/C++/Lean non-uniformity witness.

12 Retractions that affect interpretation

The following historical conclusions are explicitly excluded:

- the Möbius/M4D proof of the mixed/off-diagonal fluctuation bound;
- the one-sided mixed-class count that omitted one ordered diagonal/off class;
- a shortcut from generic random-matrix universality to this structured deterministic family without verified hypotheses;
- the unrestricted twisted-Jacobi formula that omitted its domain condition;
- the all-orders arcsine/Catalan conclusion based on the unresolved word-trace rewrite;
- unsupported general $\mathrm{Sp}(2g)$ extrapolations from finite diagnostics;
- finite numerical GOE "passes" interpreted as proof of a limiting law;
- the temporary entrywise-real Hermitian projection used during repository reconstruction.

The failure mechanisms are preserved in the research record so they remain useful regression tests.

13 Conclusion

The reconstruction leaves a narrower but substantially stronger result than the historical workspace claimed. The pair-key bijection, exact $(K-1)(K-2)$ constrained count, weighted diagonal, fourth-moment coupling, Fourier mean/variance reduction, Cayley transform, complete 720-matching audit, the explicit E_{11} and E_{13} resultant sets, and the $(461, 23)$

obstruction are reproducible. Earlier finite torus, scale and twisted-Jacobi results survive at their actual finite scope, while numerical spectral records are now regenerated from the correct complex Hermitian matrix.

The main analytic task is sharply localized: convert the historical elliptic organization into a publication-grade family and verify the exact sheaf and isogeny hypotheses required for square-root correlation. Even after that, microscopic BGS statistics require a separate local-law bridge.

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